

# Saravanakumar U

Full Stack Software Developer

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## Professional Summary

Full Stack Software Developer skilled in building scalable and efficient software solutions across frontend and backend development. Experienced in designing robust APIs using Python (FastAPI), developing responsive UIs with React.js, and integrating modern databases like Trino and Iceberg. Proficient in DevOps tools including Docker, Kubernetes, and Helm, with experience in cloud-native deployments using OpenShift. Adept at creating real-time observability platforms, automating workflows with GitHub Actions, and applying AI/ML models for anomaly detection. Passionate about delivering high-quality, secure, and maintainable applications through microservices architecture, with a strong focus on performance optimization and user experience.

## Work Experience

### Full Stack Software Developer

Zaga Open Source Pvt Ltd | Apr 2024 - Present | Kovilpatti, India

- Designed and implemented a **Data Mesh Architecture** utilizing **OpenTelemetry (OTel), Kafka, Hive, Iceberg, MinIO, Trino, and Redis** for scalable and efficient data processing.
- Developed a **real-time observability platform** by integrating **tracing, logging, and metrics collection** to monitor system health.
- Built a high-performance **data ingestion framework** using **FastAPI**, enabling seamless data flow between microservices.
- Engineered a **React-based frontend** to visualize infrastructure monitoring, system health, and analytics dashboards.
- Implemented **role-based access control (RBAC)** for data access management, leveraging **Trino and Iceberg** for secure and structured access policies.
- Led **Kubernetes, Docker, Helm, and OpenShift deployments**, automating application provisioning and ensuring **high availability and scalability**.
- Optimized **MinIO-based object storage** for handling large-scale telemetry and analytics data, improving storage efficiency and retrieval performance.

- Developed real-time **data analytics and processing pipelines** using Kafka and Trino, enabling fast query execution for observability insights.
- Developed and fine-tuned **AI/ML models** for anomaly detection using **Isolation Forest and Hugging Face**.
- Focused on **performance tuning, monitoring, and optimization** to ensure seamless user experience and efficient system operations.

## Software Development Engineer Intern

Icanio Technologies | Sep 2023 - Feb 2024 | Tirunelveli, India

- Developed user interfaces and components using React.js, ensuring responsiveness and a seamless user experience across different devices and browsers.
  - Contributed to backend development by building RESTful APIs, handling authentication, and optimizing database operations using FastAPI, MongoDB, and PostgreSQL.
  - Worked closely with designers and backend developers to integrate frontend components with backend services.
  - Actively participated in code reviews, team meetings, and adopted agile methodologies.
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## Projects

### Online Meat Delivery App

- Designed and implemented a microservices-based architecture using **FastAPI** for modularity and scalability.
- Developed backend services, including **User, Product, Order, Delivery, and Authentication** services.
- Implemented **JWT-based authentication** and **role-based access control** for secure API access.
- Built the **customer mobile app using Flutter** with features like product browsing, order placement, and real-time tracking.
- Created a **delivery partner app using React Native** to manage pickups, delivery routes, and status updates.
- Developed a **React.js admin dashboard** for managing products, orders, delivery tracking, and customer support.
- Integrated **payment gateway** Razorpay for secure online transactions.
- Enabled **real-time order updates and delivery tracking** using WebSocket and Redis.
- Containerized services using **Docker** and prepared for deployment on **OpenShift/Kubernetes**.
- Used **PostgreSQL** for storing user data, product inventory, and transaction history.
- Implemented **CI/CD pipelines** using GitHub Actions for automated build and deployment.

- Ensured responsive, user-friendly UI with smooth navigation and error handling across all apps.
- Integrated **push notifications** to alert users about order status, offers, and delivery updates.
- Focused on performance optimization and secure communication between services.

## Data Lake Management & Data Product Platform using Apache Iceberg, Python, and Trino

Designed and developed a backend-driven data management platform that enables organizations to create, manage, and publish structured data products on top of a data lake. The application integrates **Apache Iceberg (REST backend)** for schema/catalog/table management and a custom **Python backend** for business-oriented data product workflows.

Key Features:

- Created logical data structures (domains, catalogs, schemas) through a user-friendly UI.
- Leveraged Apache Iceberg and MinIO for scalable table storage and versioned data management.
- Built a Python-based backend to manage data products and expose queries as APIs.
- Integrated **Trino** as a query engine, enabling users to explore data, build queries visually, and apply filters without writing code.
- Enabled publishing of saved queries as REST APIs to be consumed by other teams or applications.

**Technologies Used:** Python, FastAPI, Docker, Apache Iceberg, MinIO, Trino, PostgreSQL, REST APIs

## Application Performance Monitoring (APM) System

- Implemented a robust monitoring system for tracing, logging, and collecting metrics to enhance real-time observability and debugging.
- Integrated OpenTelemetry (OTel) to collect telemetry data from various microservices and applications.
- Provided deep insights into application performance, helping teams optimize and troubleshoot issues efficiently.

## Infrastructure Monitoring Platform

- Designed a system to monitor nodes, pods, events, and cluster health across Kubernetes environments.
- Developed real-time dashboards using Apache Superset for visualizing infrastructure health and usage patterns.
- Integrated Prometheus and Grafana for real-time alerting and performance analytics.

## **Edge Monitoring Solution**

- Developed an edge monitoring framework to connect multiple edge devices, enabling visibility of pod, node, and cluster information in a unified dashboard.
- Ensured seamless integration with existing observability platforms to provide real-time insights into edge infrastructure.
- Implemented lightweight monitoring agents on edge devices to capture key performance metrics.

## **Sustainability Monitoring Dashboard**

- Created a sustainability-focused dashboard for monitoring the resource efficiency of Kubernetes clusters, including node and pod sustainability tracking.
- Optimized resource allocation and sustainability by analyzing historical and real-time data on cluster resource consumption.
- Enhanced cloud cost efficiency by implementing intelligent scaling recommendations.

## **VPBank - Vietnam Role-Based Access Control**

- Developed a role-based access control system using React for the frontend and Python FastAPI for backend API services.
  - Implemented role-specific access restrictions for Admin, Gold, and Platinum customers.
  - Utilized Trino with file-based access control to manage schema and table-level access based on user roles.
  - Enabled dynamic permissions for schema and catalog table management.
  - Integrated customer loyalty data to enable promotional program targeting based on spending behavior.
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## **Education**

**Bachelor of Engineering in Electronics and Communication Engineering**

## Skills

- **Programming Languages:** JavaScript, TypeScript, Python
  - **Frontend:** React.js, Next.js, Redux, Tailwind CSS
  - **Backend:** FastAPI, Node.js, Nest.js, Firebase, Express JS, REST APIs, Websocket, Redis, Kafka
  - **Python:** FastAPI, Pytest, PySpark, Pyrinio, Pylceberg,pyjwt, pandas
  - **Databases:** SQL, MongoDB, Trino, Iceberg
  - **Mobile App development:** Flutter, React Native, Expo
  - **DevOps & Tools:** GitHub Actions, Docker, OpenShift, Kubernetes, Helm, MinIO, CI/CD Pipeline
  - **Big Data & Analytics:** Apache Spark, Apache Iceberg, Apache Kafka, Apache Hive, Apache Superset
  - **Observability & Monitoring:** OpenTelemetry (OTel), Prometheus, Grafana
  - **AI/ML & NLP:** YOLO, Isolation Forest, Hugging Face, GPT-2 fine-tuning
  - **Vector Databases & RAG:** Milvus, Retrieval-Augmented Generation (RAG) pipeline
  - **Third Party Integration:** Razorpay,Google Maps
  - **OS:** Linux – RHEL, Windows
  - **Others:** Data Mesh, Observability, Role-Based Access Control
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## Interests

- Web Development
  - Microservices Architecture
  - Observability & Monitoring
  - Open Source Contributions
  - DevOps & Automation
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